

Malawi Malaria Burden Stratification Workshop

8–9 April 2026: Sigelege Hotel, Salima, Malawi

Presented by

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Workshop objectives and process

Objectives of the workshop

1. Review the latest malaria burden evidence to inform district-level stratification.
2. Validate the indicators, cut-offs, and analytical methods used in the stratification process.
3. Reach consensus on the final harmonized malaria burden strata for Malawi.
4. Use the stratification findings to inform tailored intervention planning, targeting, and resource allocation.

Core process followed

- 1 Review of previous stratification processes
- 2 Review of data inputs and processing
- 3 Stratification process and scenario comparison
- 4 Agreement on final strata and next actions

Evolution of the Malaria Stratification Process in Malawi

2021

Initial stratification exercise that informed the Global Fund application and the 2022–2023 Strategic Plan



2022–2023

Strengthened the methodology by improving the analytical process and incorporating adjustments for important data quality parameters, including test positivity, treatment-seeking, and reporting rates.

2024–2025(current)

Applied the refined process to support the Global Fund Cycle 8 application and the mid-term review of the Strategic Plan

Evolution of the Malaria Stratification Process in Malawi

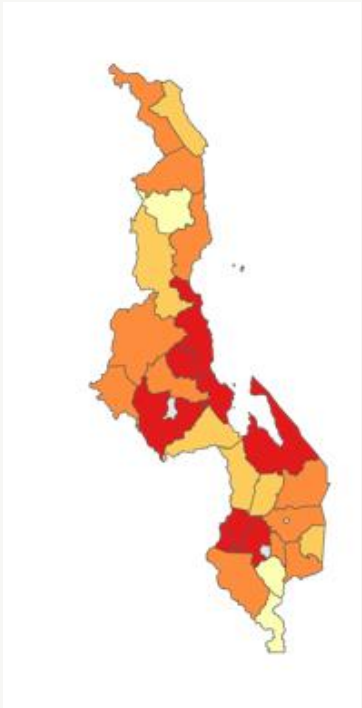
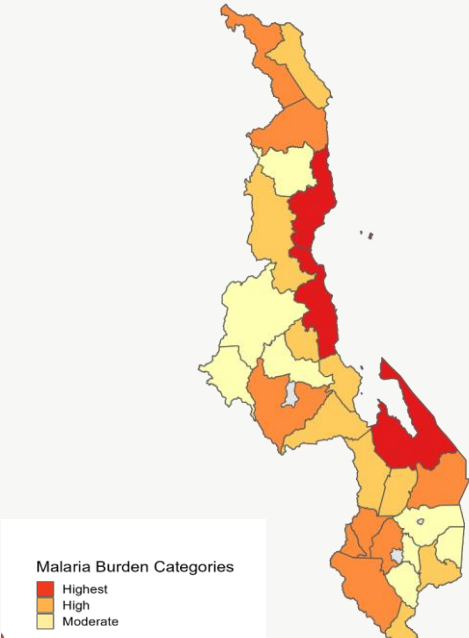
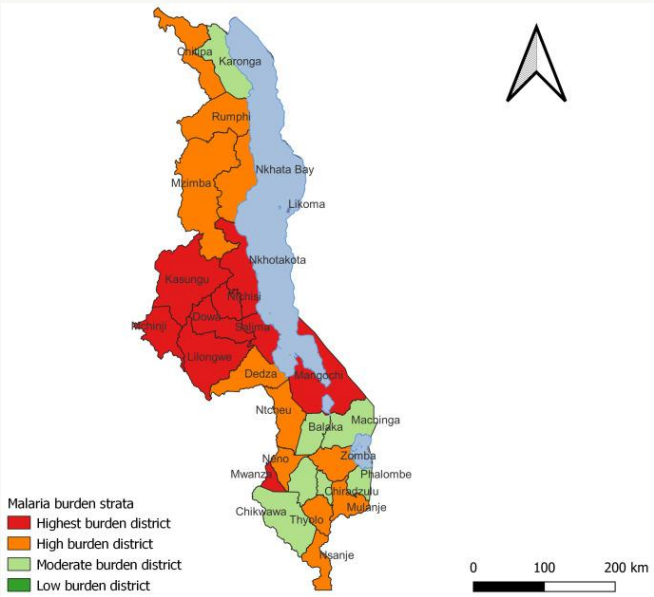
Districts

Year	Highest Burden	High Burden	Moderate Burden	Low Burden
2021	10	11	7	0
2022	4	8	9	8
2023	8	12	6	3
2024	?	?	?	?
2025	?	?	?	?

2021

2022

2023

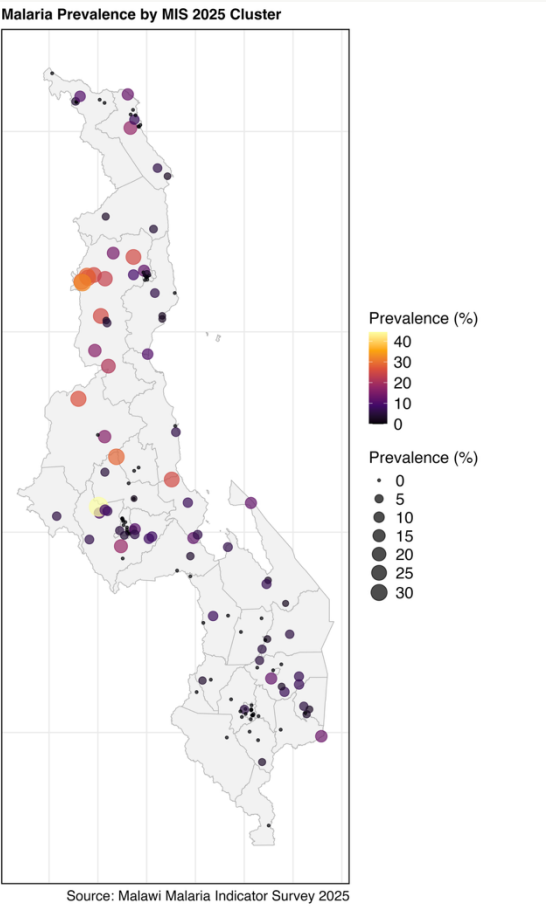


Data inputs- Incidence and mortality

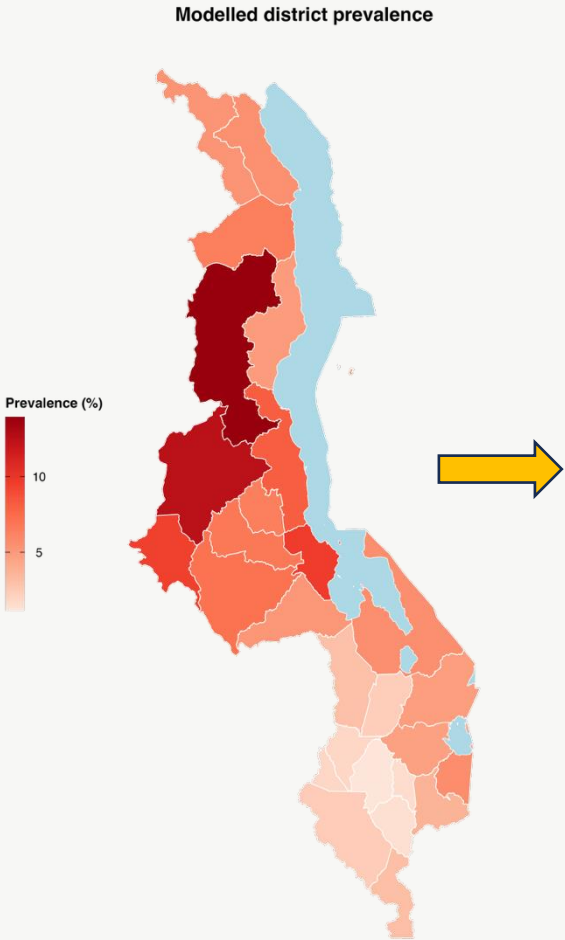
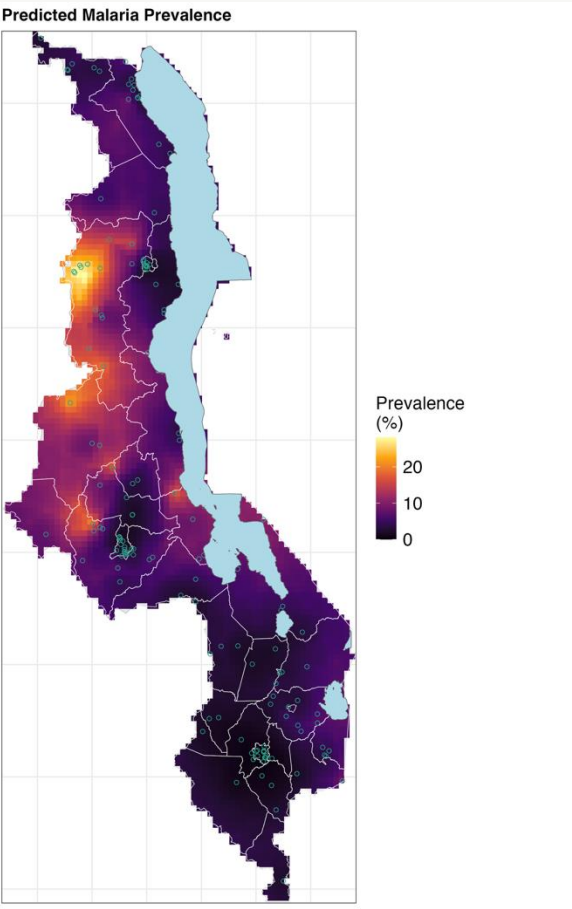
District	Mortality 2025	Cases 2025	Final adjusted 2025* Adj. incidence 2025	Mortality 2024	Cases 2024	Final adjusted 2024* Adj. incidence 2024		
Balaka	49	60,492	128,726	0.242	66	135,517	63,696	0.123
Blantyre	7	201,353	397,785	0.278	16	436,706	204,072	0.145
Chikwawa	304	130,035	276,544	0.423	85	178,301	85,205	0.133
Chiradzulu	8	58,011	122,089	0.303	40	117,712	55,420	0.14
Chitipa	14	67,377	100,036	0.378	18	68,296	44,000	0.169
Dedza	58	239,762	554,010	0.573	80	239,023	102,814	0.108
Dowa	55	277,258	630,419	0.682	49	256,777	112,996	0.125
Karonga	38	132,489	197,801	0.468	46	174,490	116,965	0.282
Kasungu	130	677,045	1,538,341	1.547	153	537,106	236,709	0.243
Likoma	0	2,013	3,006	0.181	0	4,732	3,170	0.194
Lilongwe	214	677,343	1,554,621	0.494	378	871,130	381,732	0.124
Machinga	57	172,290	367,189	0.393	105	309,463	139,695	0.155
Mangochi	83	158,849	339,059	0.237	201	326,289	138,921	0.1
Mchinji	99	304,136	652,822	0.931	66	381,256	158,104	0.23
Mulanje	53	210,154	447,205	0.56	119	401,381	188,963	0.242
Mwanza	45	74,728	152,469	0.944	56	120,100	45,162	0.288
Mzimba Nort	2	163,484	245,264	0.182	4	142,596	94,890	0.072
Mzimba Sout	54	278,197	404,997	—	91	305,881	205,609	—
Neno	12	65,741	140,012	0.951	19	203,502	95,684	0.663
Nkhata Bay	31	115,276	167,759	0.524	33	259,420	167,335	0.532
Nkhotakota	63	115,418	259,218	0.57	99	315,038	140,389	0.315
Nsanje	46	100,722	213,142	0.632	83	126,550	59,127	0.178
Ntcheu	86	214,621	487,891	0.613	103	248,810	109,498	0.141
Ntchisi	30	213,070	482,236	1.244	60	258,762	110,443	0.293
Phalombe	10	54,816	116,633	0.226	16	110,569	51,904	0.103
Rumphi	8	99,711	146,463	0.553	36	80,624	52,623	0.203
Salima	77	159,459	363,453	0.625	170	329,999	143,031	0.253
Thyolo	5	101,432	211,281	0.263	40	183,026	86,000	0.108
Zomba	3	282,229	600,717	0.61	20	476,247	224,224	0.232
* Adjusted for reporting rate, positivity rate and treatment seeking								

Malaria cases are adjusted for total positivity rate, reporting rate and treatment seeking

Deriving District-level Prevalence



Modelling



District	Prevalence
Mzimba	13.90%
Kasungu	12.50%
Salima	9.60%
Mchinji	9.40%
Nkhotakota	8.10%
Lilongwe	7.10%
Dowa	6.80%
Rumphi	6.50%
Ntchisi	6.50%
Likoma	5.80%
Phalombe	5.60%
Mangochi	5.60%
Karonga	5.50%
Chitipa	5.30%
Dedza	5.10%
Nkhata Bay	4.90%
Machinga	4.80%
Zomba	4.60%
Mulanje	3.70%
Nsanje	3.10%
Ntcheu	3.00%
Chikwawa	2.40%
Balaka	2.30%
Mwanza	1.90%
Neno	1.90%
Chiradzulu	1.60%
Thyolo	1.40%
Blantyre	1.20%

Burden Stratification Criteria

Strata	Scenario 1: Population Adjusted Approach			Scenario 2: High Burden High Impact Approach		
	Incidence	Prevalence	Malaria Mortality Rate (Under 5)	Total Cases	Prevalence	Malaria Mortality Totals (Under 5)
Highest Burden Cutoff	>500 cases/1000	>0.2 PfPr 2-10	> 70 deaths per 100K	>300,000 cases	>0.2 PfPr 2-10	> 150 total deaths
High Burden Cutoff	200-499 cases/1000	0.1-0.2 PfPr 2-10	50-70 deaths per 100K	200,000-300,000 cases	0.1-0.2 PfPr 2-10	100-150 total deaths
Moderate Burden Cutoff	<200 cases/1000	<0.1 PfPr 2-10	<50 deaths per 100K	< 200,000 cases	<0.1 PfPr 2-10	<100 total deaths

Selection of Stratification Criteria

Process

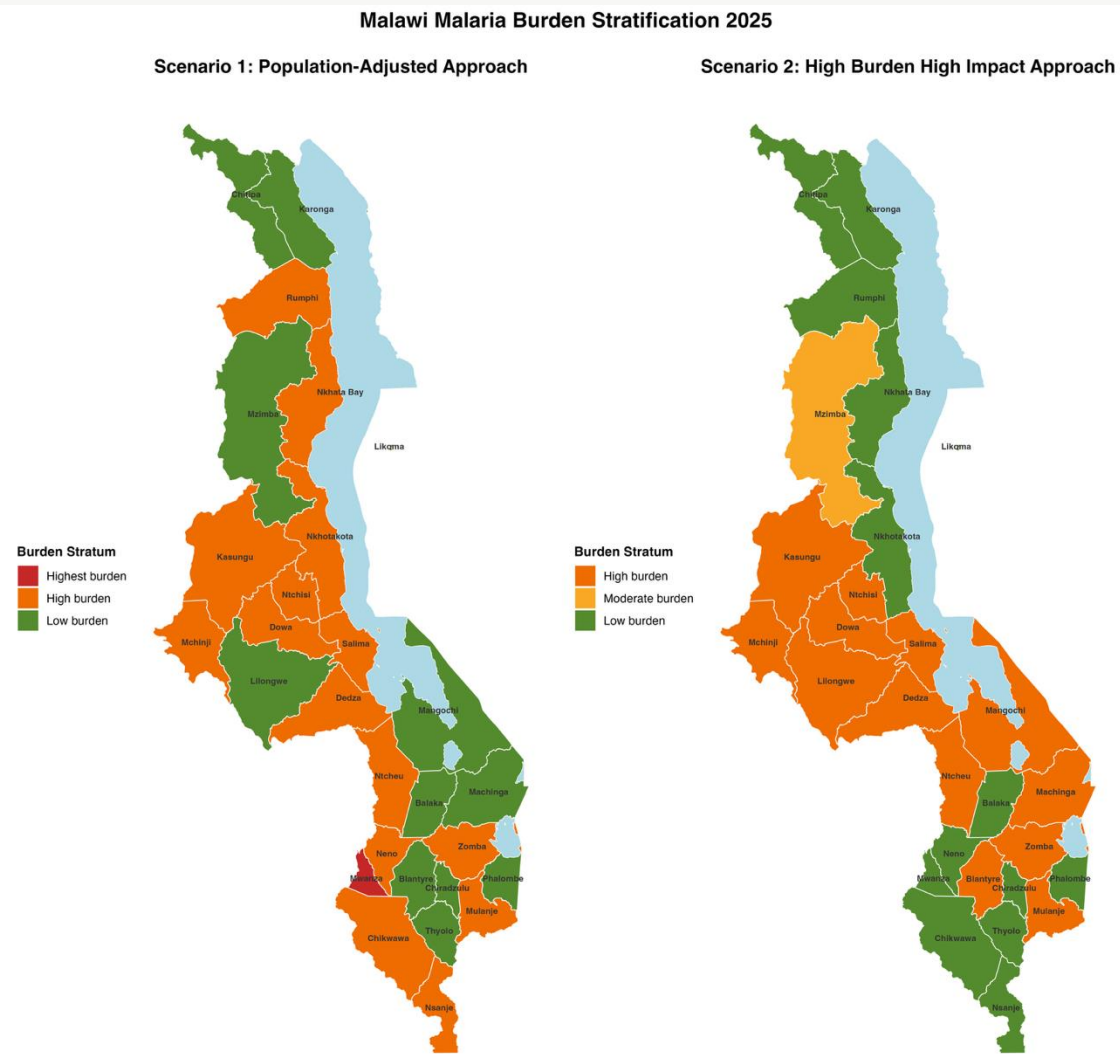
- Selected malaria burden input statistics based on roundtable discussion and data availability
- Selected malaria burden category cut-offs using combination of data-based decision making and historical precedent
 - Data-based decisions made by sorting the data into equal weighted quantiles and adjusting to rounded numbers when necessary
 - Prevalence cutoffs established based on previous WHO classifications
- Consensus required before any decisions were finalized

Strata level	Definition
Highest burden strata	At least 2 of the highest-burden cutoffs
High burden strata	Only 1 of the highest-burden <u>cutoffs</u>
Moderate burden strata	At least 2 of the high-burden cutoffs and 0 highest-burden cutoffs
Low burden strata	< 2 of the high-burden <u>cutoffs</u> , at least 1 of the moderate burden cut offs and 0 of the highest-burden

Stratification scenario results- 2025

Strata level	Scenario 1	Scenario 2	Final
Highest	Mwanza	None	Mwanza
High	Chikwawa, Dedza, Dowa, Kasungu, Mchinji, Mulanje, Neno, Nkhatabay, Nkhotakota, Nsanje, Ntcheu, Ntchisi, Rumphu, Salima, Zomba	Blantyre, Dedza, Dowa, Kasungu, Lilongwe, Machinga, Mangochi, Mchinji, Mulanje, Ntcheu, Ntchisi, Salima, Zomba	Blantyre, Chikwawa, Dedza, Dowa, Kasungu, Lilongwe, Machinga, Mangochi, Mchinji, Mulanje, Neno, Nkhatabay, Nkhotakota, Nsanje, Ntcheu, Ntchisi, Rumphu, Salima, Zomba
Moderate	None	Mzimba	Mzimba
Low	Balaka, Blantyre, Chiradzulu, Chitipa, Karonga, Likoma, Lilongwe, Machinga, Mangochi, Mzimba, Phalombe, Thyolo	Mwanza, Chikwawa, Neno, Nkhatabay, Nkhotakota, Nsanje, Rumphu, Balaka, Chiradzulu, Chitipa, Karonga, Likoma, Phalombe, Thyolo	Balaka, Chiradzulu, Chitipa, Karonga, Likoma, Phalombe, Thyolo

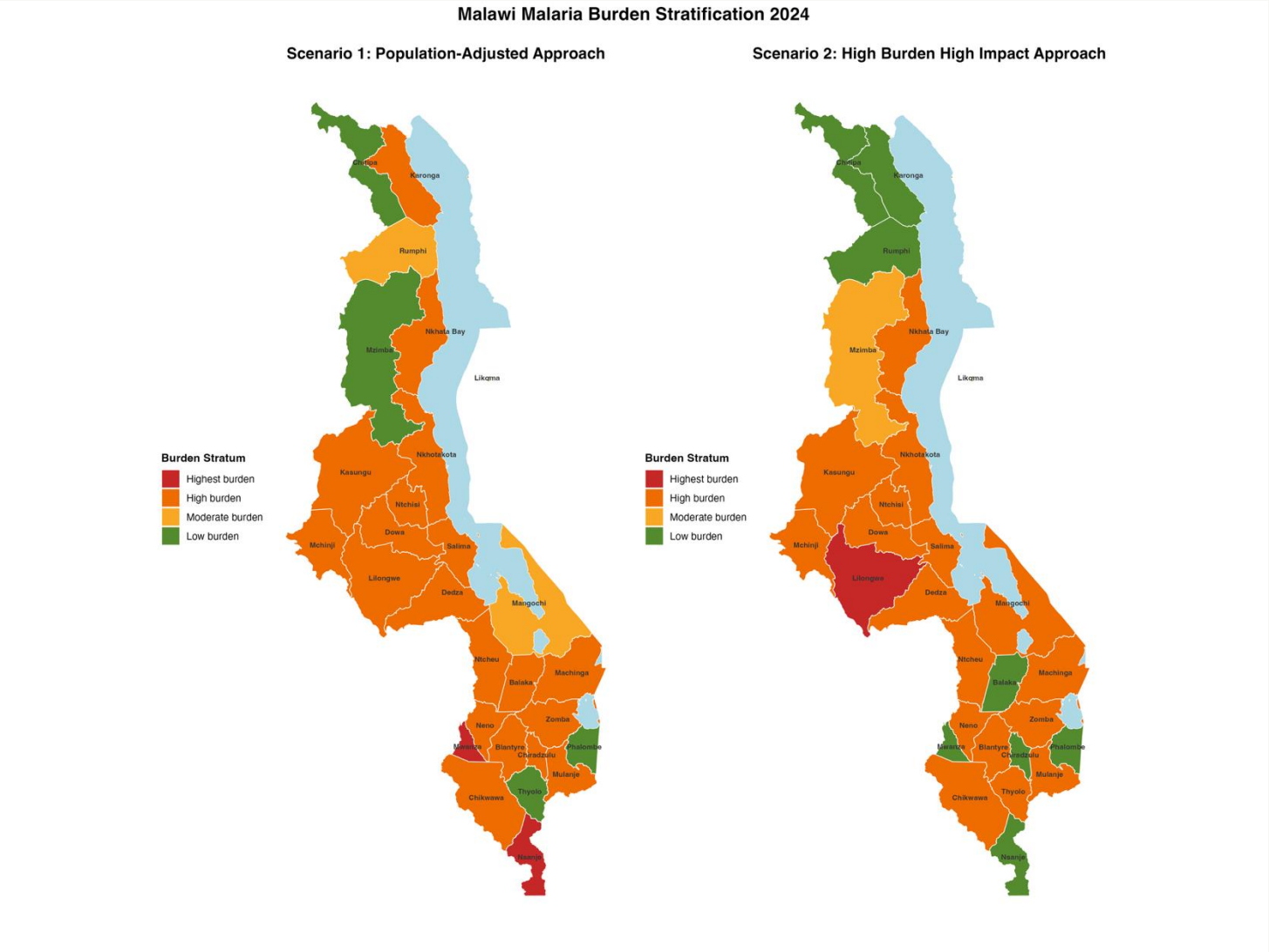
Stratification scenario results- 2025



Stratification scenario results- 2024

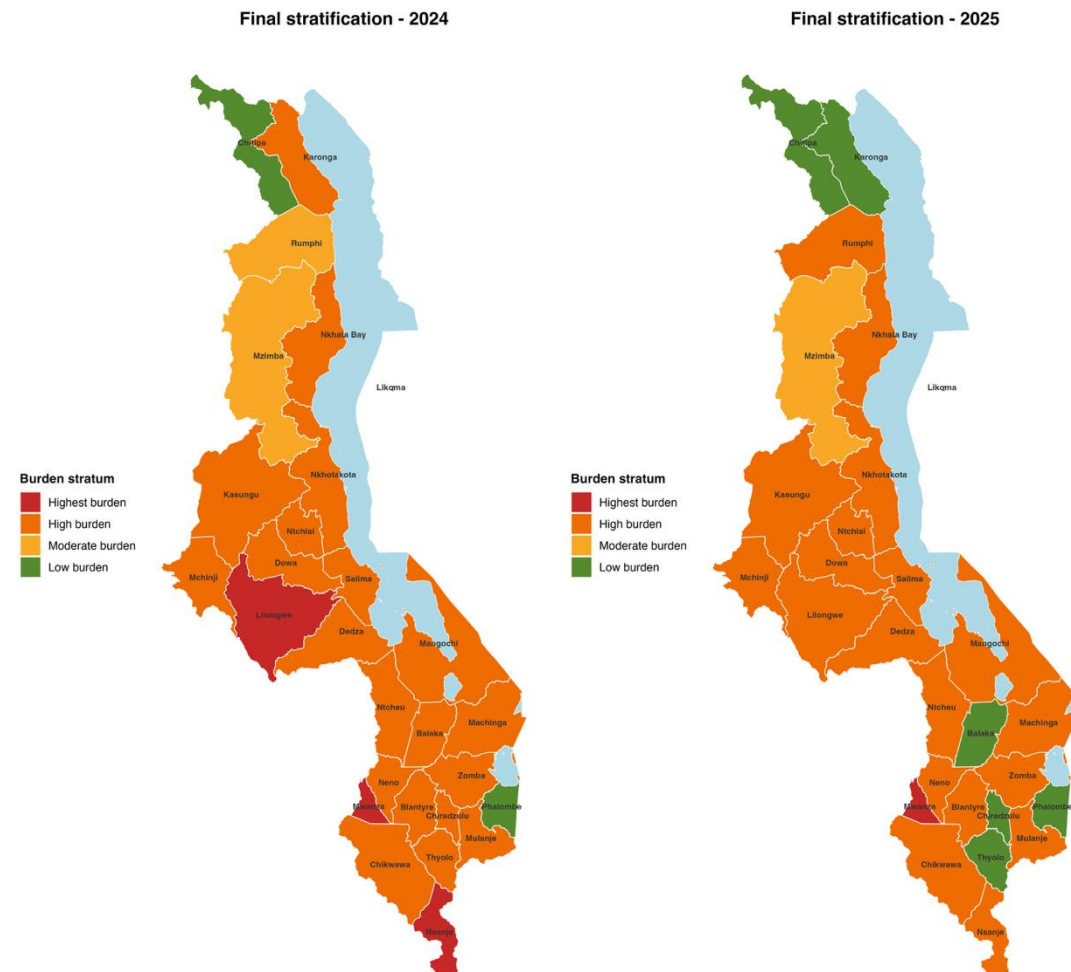
Strata level	Scenario 1	Scenario 2	Final
Highest	Mwanza, Nsanje	Lilongwe	Mwanza, Nsanje, Lilongwe
High	Balaka, Blantyre, Chikwawa, Chiradzulu, Dedza, Dowa, Karonga, Kasungu, Lilongwe, Machinga, Mchinji, Mulanje, Neno, Nkhata Bay, Nkhatakota, Ntcheu, Ntchisi, Salima, Zomba	Blantyre, Chikwawa, Dedza, Dowa, Kasungu, Machinga, Mchinji, Mulanje, Neno, Nkhata Bay, Nkhatakota, Ntcheu, Ntchisi, Salima, Zomba, Mangochi, Thyolo	Balaka, Blantyre, Chikwawa, Chiradzulu, Dedza, Dowa, Karonga, Kasungu, Machinga, Mchinji, Mulanje, Neno, Nkhata Bay, Nkhatakota, Ntcheu, Ntchisi, Salima, Zomba, Mangochi, Thyolo
Moderate	Mangochi, Rumphi	Mzimba	Rumphi, Mzimba
Low	Chitipa, Likoma, Mzimba, Phalombe, Thyolo	Mwanza, Nsanje, Balaka, Chiradzulu, Karonga, Rumphi, Chitipa, Likoma, Phalombe	Chitipa, Likoma, Phalombe

Stratification scenario results- 2024



Stratification results- Final

Strata level	Final 2024	Final 2025
Highest	Mwanza, Nsanje, Lilongwe	Mwanza
High	Balaka, Blantyre, Chikwawa, Chiradzulu, Dedza, Dowa, Karonga, Kasungu, Machinga, Mangochi, Mchinji, Mulanje, Neno, Nkhata Bay, Nkhotakota, Nsanje, Ntcheu, Ntchisi, Rumphi, Salima, Thyolo, Zomba	Blantyre, Chikwawa, Dedza, Dowa, Kasungu, Lilongwe, Machinga, Mangochi, Mchinji, Mulanje, Neno, Nkhata Bay, Nkhotakota, Nsanje, Ntcheu, Ntchisi, Rumphi, Salima, Zomba
Moderate	Rumphi, Mzimba	Mzimba
Low	Chitipa, Likoma, Phalombe	Balaka, Chiradzulu, Chitipa, Karonga, Likoma, Phalombe, Thyolo



Evolution of the Malaria Stratification Process in Malawi

Districts

Year	Highest Burden	High Burden	Moderate Burden	Low Burden
2021	10	11	7	0
2022	4	8	9	8
2023	8	12	6	3
2024	3	20	2	3
2025	1	19	1	7

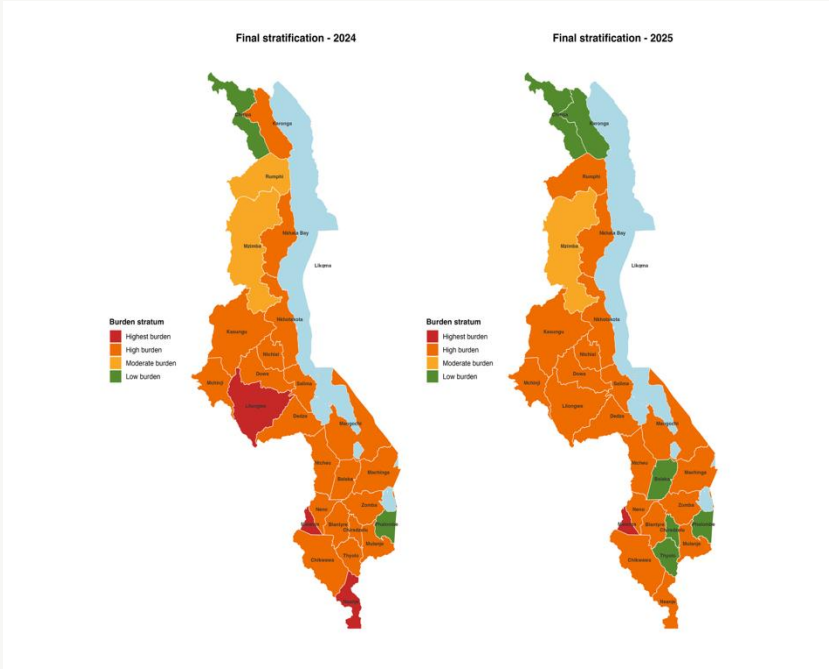
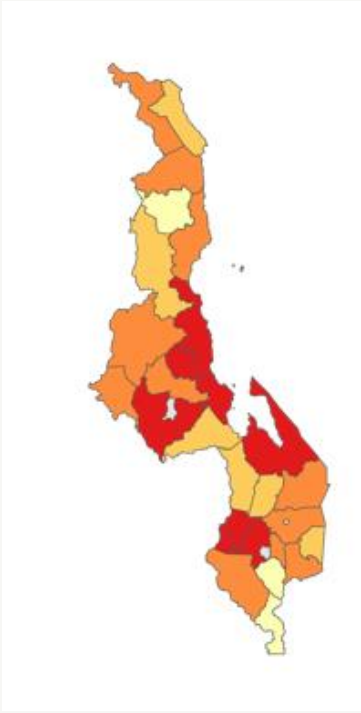
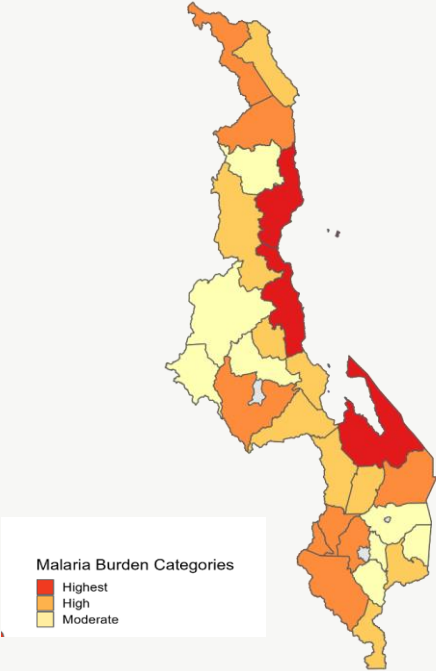
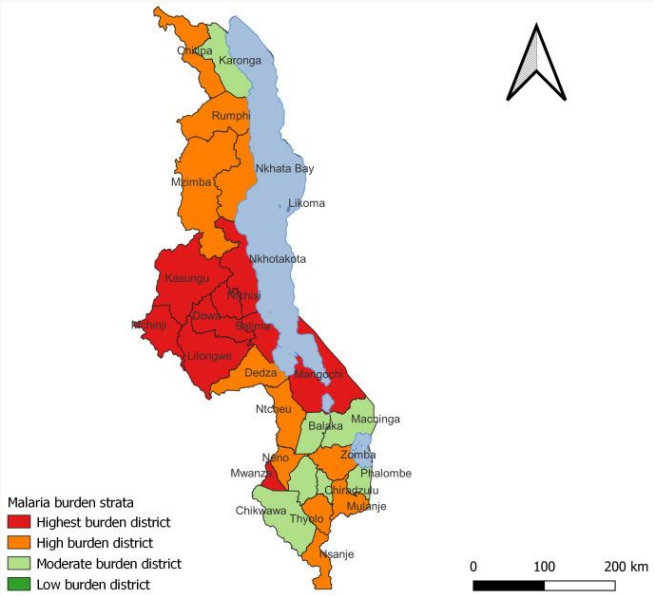
2021

2022

2023

2024

2025



Malaria Interventions

Current

- Vector control
 - Mass net distribution
 - Routine net distribution
 - IRS
- Case Management
 - Effective malaria diagnosis
 - Clinical case management with effective drugs
 - “learners treatment kit”
- Preventative treatment
 - IPTp
 - “Malaria Vaccine”
 - PDMC
- Social and behaviour change
- Surveillance monitoring and evaluation and operational research

Potential/Upcoming

- Vector control
 - School net distribution
 - IRS
 - Larva source management
 - DDT
 - Genetically modified mosquitos

Case Management

- Low dose primaquine
- Scale up of learners' treatment kit”
- Preventative treatment
 - SMC
 - Malaria monoclonal antibodies
 - IPTi + in schools

Strata Category	Intervention Recommendations
Highest Burden	SMEOR(Rx m and E, TES), SBC (Interpersonal communication), Case management (ICCM, eMCCM Treatment kit), IPTp, Vector control(Nets (Routine, Mass campaign, school based), IRS, GM, DDT), Malaria Vaccine, Increase supportive supervision frequency, identify and mitigate issues from previous supervisions,
High burden	SMEOR(Rx m and E, TES), ,SBC, Case management(ICCM,eMCCM),IPTp, Vector control(Nets(Routine,Mass campaign, school based, IRS,GM, DDT),Malaria Vaccine
Moderate	SMEOR(Rx m and E, TES), , SBC, Case management,IPTp, Vector control(LSM, Nets(Routine)
Low	SMEOR(Reactive case detection, Early Warning Systems and Epidemic Response), SBC(community awareness), Case management,IPTp, Vector control((LSM, Nets(Routine)

Discussion and next steps

Finalization of subnational tailoring
Micro stratification of some districts
Understanding of drivers of burden and transmission

Acknowledgements



BILL & MELINDA
GATES *foundation*

